

Security & Compliance

VM Migration Best Practices

How hyper2kvm ensures data protection, regulatory compliance, and security hardening throughout the migration lifecycle — HIPAA, SOC 2, GDPR, FedRAMP.

Zero data exfiltration | Air-gap capable | Audit trail

Data Protection

Your VM data never leaves your control during migration.

No Cloud Upload

hyper2kvm runs entirely on-premises. Disk images are converted locally. No data sent to external services, APIs, or analytics platforms. Zero telemetry.

Encryption at Rest

Output qcow2 images support LUKS encryption. Source VMDKs with encrypted vSAN/VMFS are handled natively. Keys never leave your infrastructure.

Encryption in Transit

vSphere exports use TLS 1.2+. SSH-based remote exports use ed25519/RSA keys. KubeVirt uploads via HTTPS. No plaintext disk transfers.

Secure Cleanup

Temporary files in /tmp auto-deleted after conversion. Working directories cleaned on completion. No residual disk data left behind.

Air-Gapped Migration

For classified, SCIF, or disconnected environments: hyper2kvm works fully offline. Install via RPM/DEB from USB. Export VMDKs to external drive. Convert on isolated network. No internet required at any stage.

Regulatory Compliance

How hyper2kvm aligns with major compliance frameworks.

Framework	Requirement	hyper2kvm Compliance
HIPAA	PHI must remain in controlled environment	All processing local, no cloud dependency
HIPAA	Audit trail for data access	Structured JSON logging with trace IDs
HIPAA	Encryption of PHI at rest	LUKS-encrypted qcow2 output supported
SOC 2	Access controls	sudo-only execution, RBAC in hypersdk
SOC 2	Change management audit	YAML configs are version-controlled, reproducible
GDPR	Data residency (EU data stays in EU)	On-prem processing, no cross-border transfer
GDPR	Right to erasure	Delete source + converted images, no copies retained
FedRAMP	FIPS 140-2 crypto	Uses system OpenSSL, FIPS mode honored
PCI DSS	Network segmentation	Works on isolated networks, no external calls
ISO 27001	Information security controls	Audit logging, access control, encryption

Security Hardening

What hyper2kvm does to harden migrated VMs.

Credential Handling

vCenter passwords via env vars or vault references. Never stored in YAML.

vc_password_env indirection prevents credential leaks in configs.

SELinux Relabel

Triggers autorelabel on first boot after migration. Ensures all file contexts are correct for the new KVM environment. No permissive mode shortcuts.

VMware Tools Removal

Removes open-vm-tools, VMware agents, and Hyper-V integration services. Eliminates unnecessary attack surface from source hypervisor.

Driver Hardening

Injects only required virtio drivers (blk, net, scsi). No unnecessary kernel modules loaded. Minimal attack surface on target.

Audit Trail

Every migration logged with structured JSON: timestamps, trace IDs, source/target paths, user, exit codes. Ship to SIEM for compliance.

Reproducible Builds

YAML configs define exact migration parameters. Same config = same result. Auditors can verify any migration was performed correctly.

Pre-Flight Security Checks

- Validates source disk integrity before conversion (extent size, descriptor parsing)
- Checks target directory permissions (no world-writable output)
- Verifies KVM/libvirt security context (SELinux labels on output images)
- Warns on insecure TLS (self-signed vCenter certs flagged in logs)
- Detects and reports Windows VMs needing VirtIO driver injection

Open Source Advantage

Why open source is more secure than proprietary migration tools.

Full Code Audit

Every line of hyper2kvm is visible on GitHub. Your security team can audit the entire codebase. No hidden backdoors, no telemetry, no phone-home.

Patch Yourself

Found a vulnerability? Fix it immediately. No waiting for vendor patches. No support ticket escalation. Fork, fix, deploy in hours.

SBOM Ready

Full dependency tree visible in go.mod and pyproject.toml. Generate SBOM for compliance. No hidden third-party components.

No Vendor Risk

Apache 2.0 license — no vendor can revoke access, change terms, or force upgrades. Your migration tool can't be rug-pulled like VMware was.